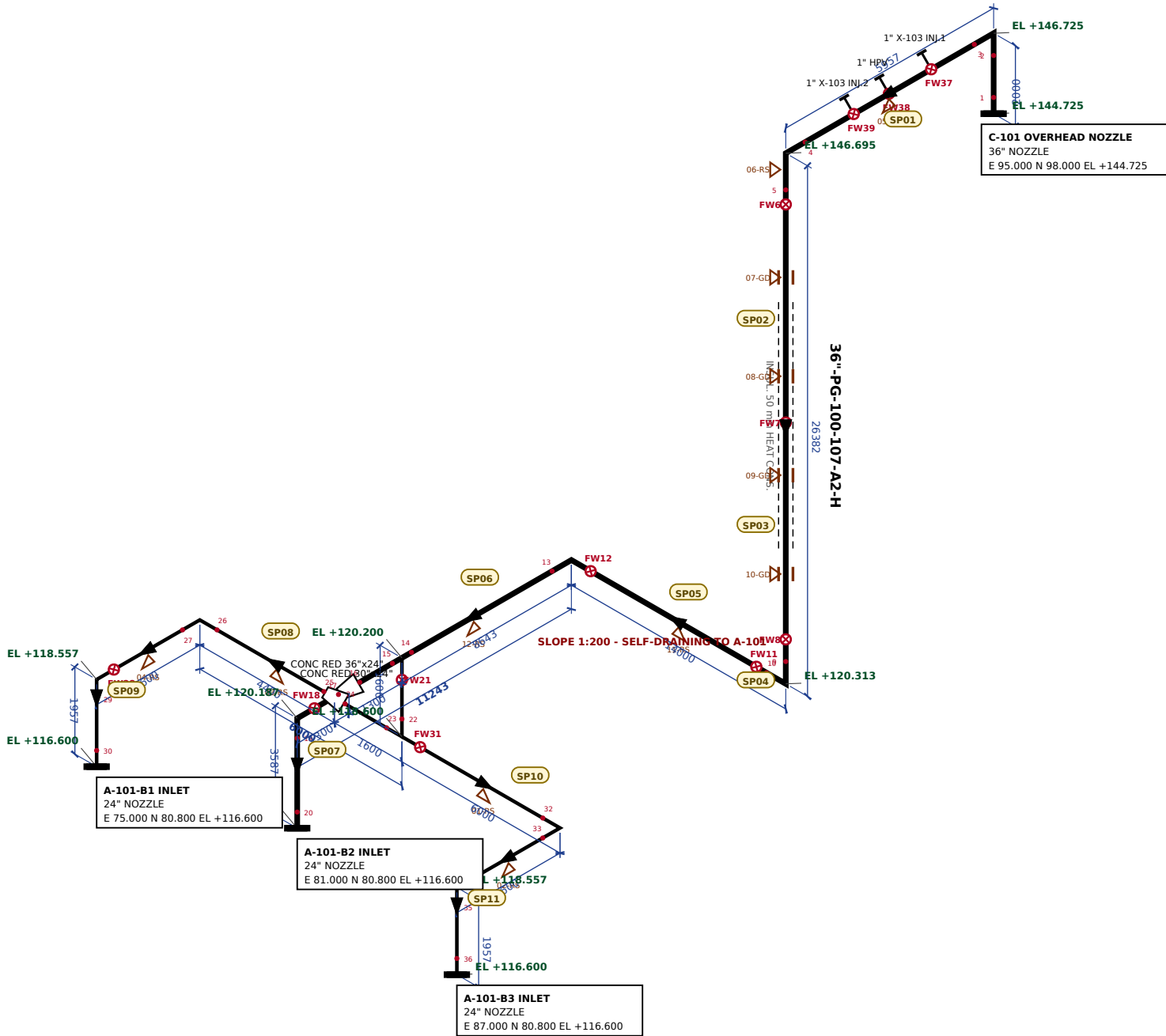
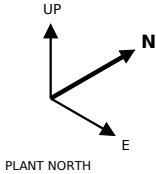


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



LINE NUMBER 36"-PG-100-107-A2-H	FROM C-101 top	TO A-101	PIPING CLASS A2 - 150# CS ASTM A106 Gr.B HIC-resistan
DESIGN P / T 3.5 barg / 165.0 C	OPER. P / T 1.19 barg / 133.8 C	TEST P (HYDRO) 5.2 barg (1.5PxST/S=1.0)	PIPE WALL STD (9.53 mm)
INSULATION H 50 mm	PWHT YES - service (sour/HIC) 595-650 C	NDE 20 % RT of BW; HB <= 200	STRESS CAT 1 - air cooler header (API 661 nozzle loads);
PG&ID CFU-100-PR-PID-007, CFU-100-PR-PID-009	FLUID / PHASE PG / V C-101 overhead vapour	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 11 spools; 26 shop / 13 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL150 B16.47A, bore STD, ASTM A105	36"	1
2	PIPE STD, BE, ASTM A672 C70 Cl.22 EFW HIC/SSC (NACE MR0103)	36"	45.5 m
3	ELBOW 90 LR, BW, STD, ASTM A234 WPB	36"	4
4	TEE RED., BW, STD, ASTM A234 WPB	36" x 30"	1
5	REDUCER CONC, BW, STD, ASTM A234 WPB	36" x 24"	1
6	PIPE STD, BE, ASTM A106 Gr.B SMLS HIC/SSC (NACE MR0103)	24"	13.6 m
7	ELBOW 90 LR, BW, STD, ASTM A234 WPB	24"	5
8	FLANGE WN RF CL150 B16.5, bore STD, ASTM A105	24"	3
9	PIPE STD, BE, ASTM A672 C70 Cl.22 EFW HIC/SSC (NACE MR0103)	30"	1.1 m
10	TEE RED., BW, STD, ASTM A234 WPB	30" x 24"	1
11	REDUCER CONC, BW, STD, ASTM A234 WPB	30" x 24"	1
12	SOCKOLET 3000# (branch), ASTM A105	1"	3
13	ERECTION: GASKET SPW 304/graphite, outer ring, CL150 B16.20	36"	1
14	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	36"	1
15	GASKET SPW 304/graphite, outer ring, CL150 B16.20	24"	3
16	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	24"	3
17	PIPE SUPPORTS (std. dwgs): guide x4, rest x8	-	12

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	36"	472	SP01-2	36"	3214	SP01-3	36"	650
SP02-1	36"	11000	SP03-1	36"	11000	SP04-1	36"	989
SP04-2	36"	650	SP05-1	36"	10607	SP06-1	36"	6599
SP06-2	36"	322	SP06-3	24"	81	SP07-1	24"	2522
SP08-1	30"	368	SP08-2	30"	736	SP08-3	24"	3181
SP08-4	24"	771	SP09-1	24"	892	SP10-1	24"	4527
SP10-2	24"	771	SP11-1	24"	892			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=10.0m SP02=11.0m SP03=11.0m SP04=4.4m SP05=10.6m SP06=11.7m SP07=4.5m SP08=8.6m SP09=2.9m SP10=7.1m
SP11=2.9m

NOTES

- Pipe/fittings/flanges per class A2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - service (sour/HIC) 595-650 C. NDE: 20 % RT of BW; HB <= 200.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Line self-draining: continuous fall 1:200 from C-101 to A-101 - no pockets.
- Inlet manifold to A-101 bays symmetrical about bay B2 (hydraulic balance, NH4Cl/HCl corrosion control).
- Neutraliser / filming amine injection quills within 1.5 m of column nozzle (X-103).
- Riser guided at C-101 platform clips; weight carried by column lugs (line & column grow together).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 36"-PG-100-107-A2-H C-101 OVERHEAD VAPOUR TO A-101		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-002		REV A